RAFFLES INSTITUTION
2021 Year 6 Preliminary Examination

Higher 2

BIOLOGY

9744/01

Paper 1 Multiple Choice

27th September 2021
1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name and shade your Index Number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions in this paper. Answer all questions. For each question there are four possible answers **A, B, C**, and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
The use of an approved calculator is expected, where appropriate.

(Erase all mistakes completely. Do not bend or fold the OMR Answer Sheet).

1. A student carried out the Benedict's test on two different types of milk, X and Y.

A sample of each type of milk was heated to 100ºC in a water-bath with Benedict's solution and the time taken for the first appearance of a colour change was recorded.

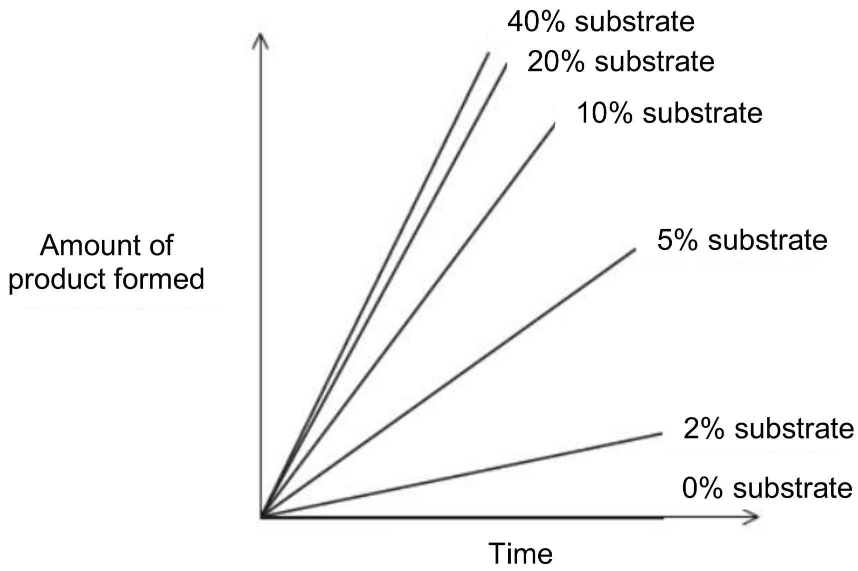
The results are shown in the table.

type of milk	time for first appearance of a colour change with Benedict's solution/s
X Y	13 26

Which row shows the biological molecule the student detected in each sample of milk and the sample of milk with the highest concentration of this biological molecule?

	biological molecule present in each sample of milk	sample of milk with the highest concentration of this biological molecule
A	glucose	X
B	glucose	Y
C	reducing sugar	X
D	reducing sugar	Y

2. The graph shows the effect of changing the substrate concentration on the early stages of an enzyme-catalysed reaction.



What can be interpreted about the rate of reaction from the graph?

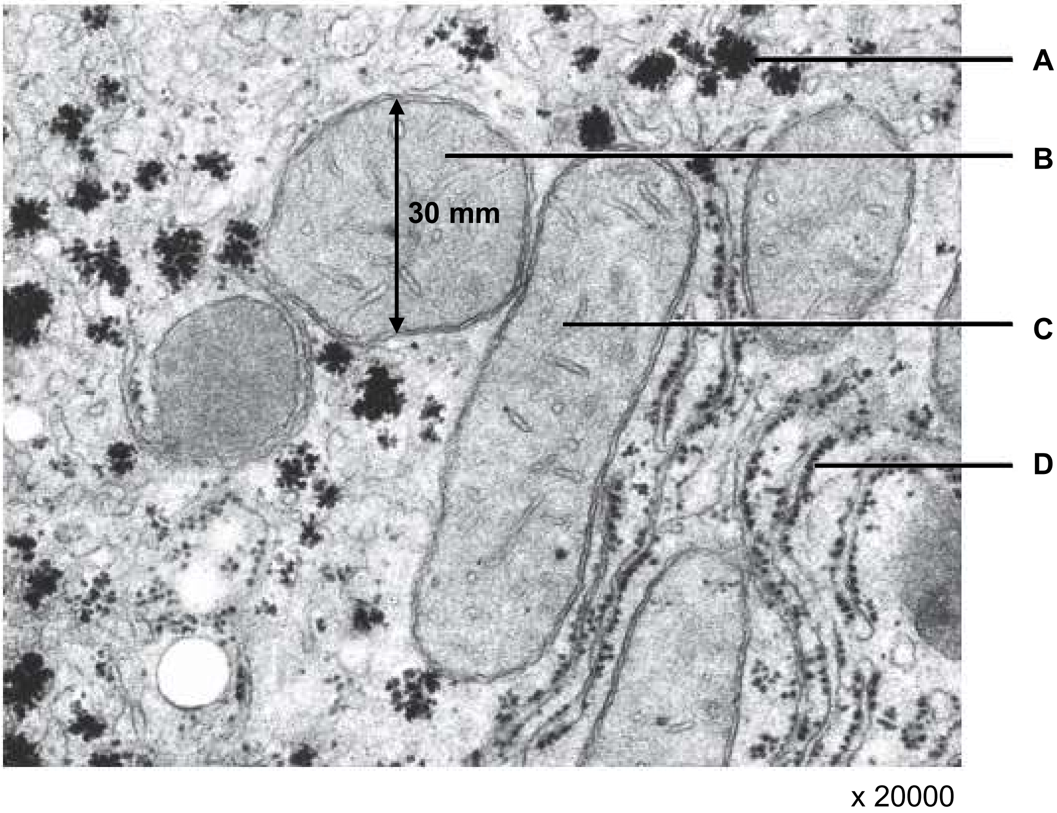
A The rate of reaction increases up to a point and then remains constant.

B The rate of reaction increases linearly with increasing substrate concentration.

C The rate of reaction increases non-linearly with increasing substrate concentration.

D The rate of reaction is not affected by any change in the substrate concentration.

3. The electron micrograph below shows some organelles and densely stained storage molecules in a mammalian liver cell.



Which of the following statements are true?

1 B and C are the same organelle.

2 Glycogen phosphorylase catalyses the formation of A from glucose.

3 The diameter of organelle B is 30mm, so its actual diameter is 1.5μm.

4 Amino acyl tRNA synthetase is synthesised at D.

A 1 and 3 only

B 2 and 3 only

C 2.and 4 only

D 1, 3 and 4 only

4. Which statement does not describe any of the membranes of the endomembrane system?

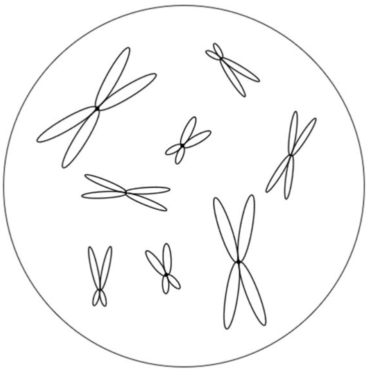
A In antigen presenting cells, MHC proteins are embedded in the membrane after synthesis.

B The membrane may possess proteins that transport pyruvate into the organelle it surrounds.

C Fluidity of the membrane allows for the secretion of proteins like insulin out of the cell and into the blood.

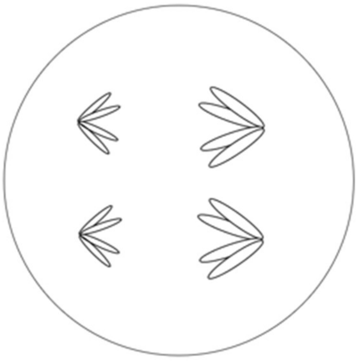
D Compartmentalisation by the membrane allows an organelle to maintain a lower pH compared to the rest of the cytosol.

5. The diagram represents the nucleus of a cell 2n = 8 in late prophase of mitosis.

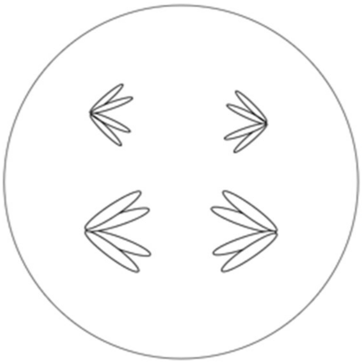


Which diagram represents a cell from the same species in anaphase II of meiosis?

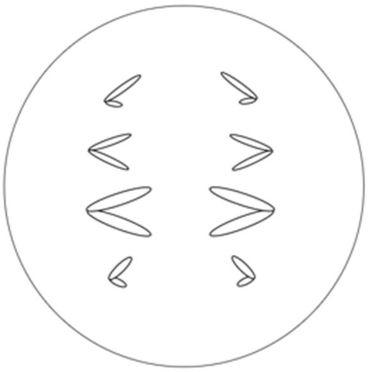
A



B

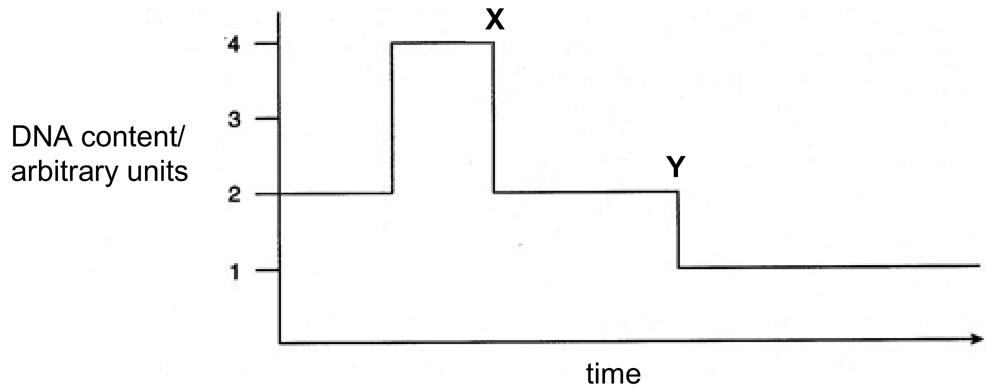


C



D

6. The graph shows the amount of DNA present in the nuclei of cells undergoing division in a multicellular organism.

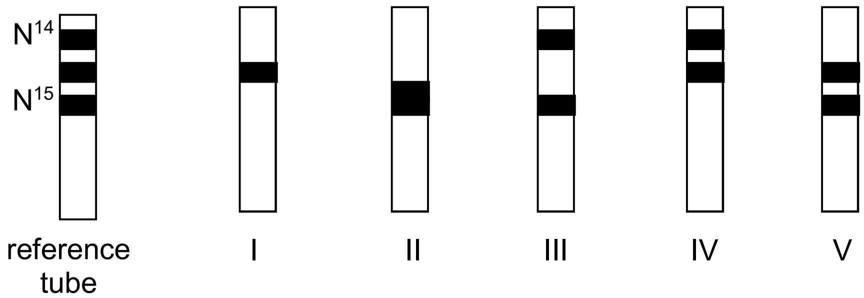


Which combination is correct?

	type of nuclear division	X	Y
A	meiosis	cytokinesis	cytokinesis
B	meiosis	telophase I	telophase II
C	mitosis	anaphase	telophase
D	mitosis	metaphase	telophase

7. Cells of the *Escherichia coli* bacterium were grown for many generations in a medium containing only the heavy isotope of nitrogen, ^{15}N. The cells were then transferred to a medium containing only ^{14}N and allowed to grow.

Samples of the bacteria were removed from the culture after the 1st round of replication and then again after the 2nd round of replication. The DNA from each sample was extracted and centrifuged at very high speed in a solution of caesium chloride.



The diagram above illustrates the DNA distribution in various centrifuge tubes, I, II, III, IV and V as proposed by a student.

Which of the above centrifuge tubes show semi-conservative DNA replication?

	tube after 1st round of replication	tube after 2nd round of replication
A	I	III
B	IV	III
C	I	IV
D	II	I

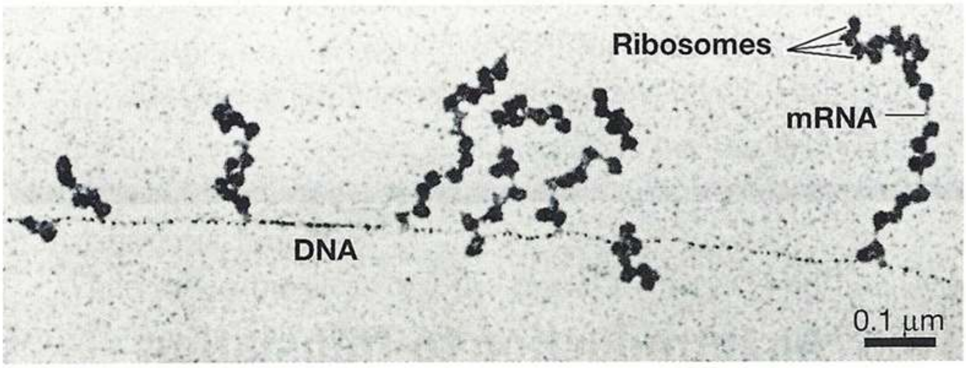
8. The mechanisms of action of five antibiotics are stated below.

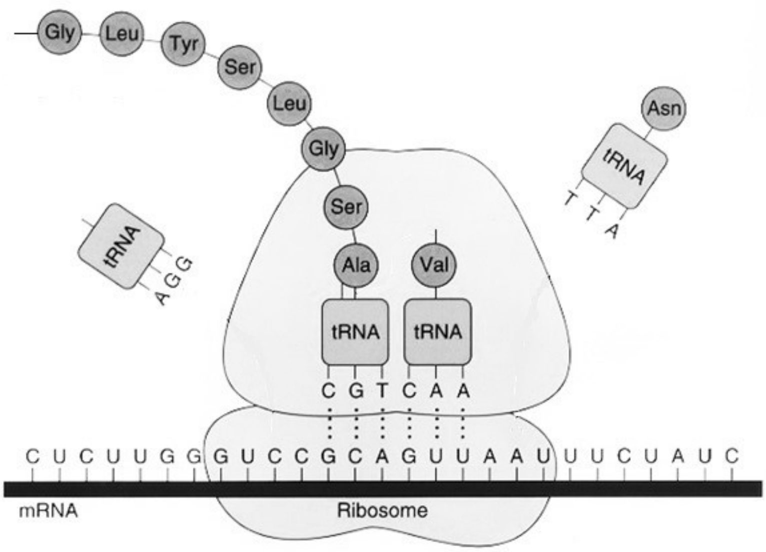
antibiotic	mechanism of action
tetracycline	binds to the 30S ribosomal subunit of a bacterium, blocking aminoacyl tRNA from binding to the A-site
clindamycin	inhibits peptidyl transferase function
rifampin	inhibits DNA-dependent RNA polymerase during transcription initiation
quinolone	inhibit the ability of topoisomerase to join nucleotide fragments
kanamycin	causes misreading of the mRNA and incorporation of incorrect amino acids

Which row correctly matches the specified mechanisms of action of these antibiotics with the direct effects on a bacteria cell?

	polypeptide with abnormal function	truncated polypeptides	no transcription	no DNA replication
A	clindamycin	kanamycin	rifampin	tetracycline
B	kanamycin	tetracycline	quinolone	clindamycin
C	kanamycin	clindamycin	rifampin	quinolone
D	clindamycin	rifampin	quinolone	tetracycline

9. The first figure below shows gene expression occurring in a cell. The second figure contains more details on the process occurring at one of the ribosomes.





Which pair of statements is true about the gene expression process shown?

I	The ribosome is translocating from right to left.
II	The information provided does not show degeneracy of the genetic code with respect to leucine.
III	These processes are occurring in the nucleus and cytoplasm of the cell.
IV	The DNA is read from the 3' to 5' direction while the mRNA is read from the 5' to 3' direction.

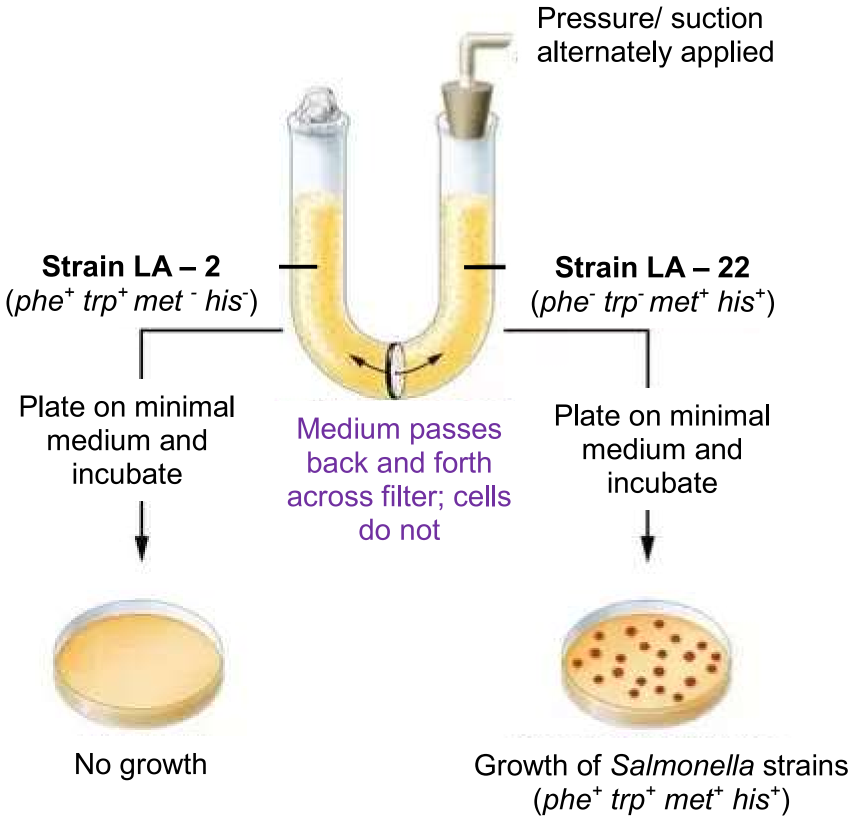
A I and III only

B I and IV only

C II and III only

D II and IV only

10. In the experiment by Zinder and Lederberg in 1952, investigating genetic exchange in *Salmonella* strains was demonstrated using a U-tube set up with a filter separating the two arms as shown in the figure below.



Strain LA-2 is able to synthesise phenylalanine and tryptophan but not methionine and histidine. Strain LA-22 is able to synthesise methionine and histidine but not phenylalanine and tryptophan.

Which of the following conclusions can be drawn from the results?

I The method of genetic transfer must be transduction.

II Direct contact between cells is not required for the transfer of genes.

III LA-2 is resistant to phage infection.

IV LA-2 is the donor of the phe$^+$ and trp$^+$ genes.

A IV only

B I and II only

C II, and IV only

D I, II and III only

11. Inducible genes code for the synthesis of enzymes involved in catabolic pathways.

Which mechanism in bacteria can be used to control the activity of inducible genes?

A Excess product of the catabolic pathway binds to the repressor to inactivate it.

B Excess substrate of the catabolic pathway binds to the operator, blocking RNA polymerase form binding to the promoter.

C Excess substrate of the catabolic pathway binds to the repressor to inactivate it.

D Excess product of the catabolic pathway binds to the repressor to activate it.

12. Which of the following statement/s on viruses is/are false?

1 The HIV polyprotein is synthesised from a polycistronic mRNA template.

2 Influenza B undergoes genetic drift as there is an accumulation of mutations in the haemagglutinin gene with time.

3 T4 phage is the agent involved in specialised transduction.

4 Replication cycles of both HIV and lambda phage have a latent phase.

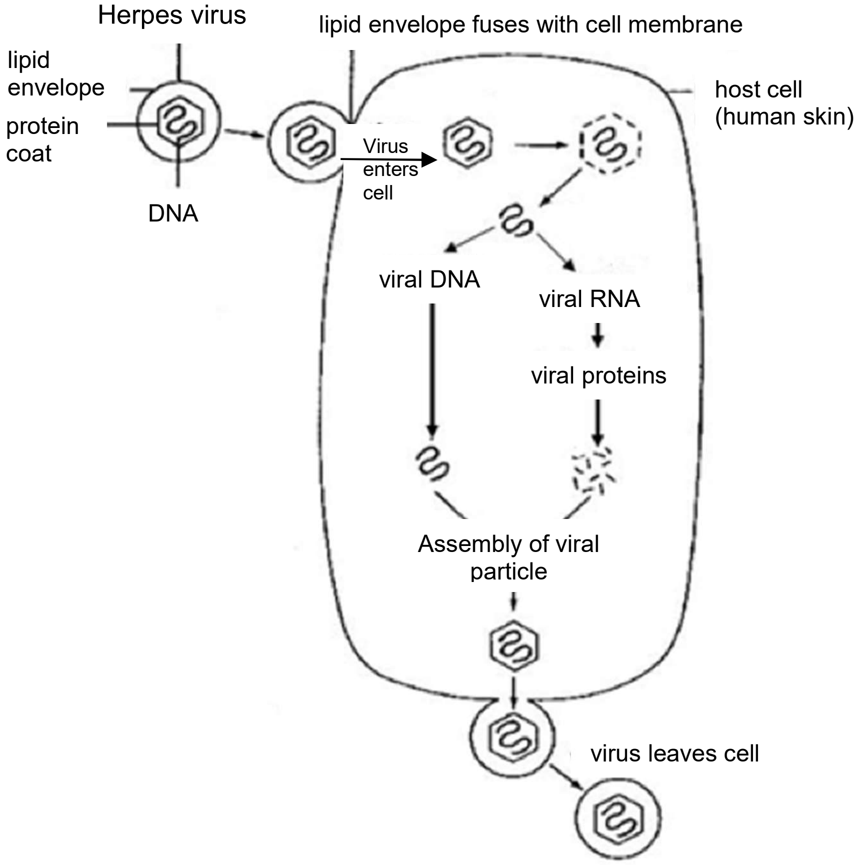
A 3 only

B 4 only

C 1 and 2 only

D 1, 2 and 3 only

13. The diagram below shows the replication cycle of the herpes virus which causes cold sores in the mouth.



Which of the following rows represents the herpes virus replication cycle?

	penetration	enzyme involved in genome replication	release
A	endocytosis	RNA dependant RNA polymerase	exocytosis
B	fusion	DNA dependant RNA polymerase	budding
C	endocytosis	reverse transcriptase	exocytosis
D	fusion	DNA polymerase	budding

14. Which of the following statement(s) is/are incorrect regarding telomeres?

1 Telomeres prevent the end-replication problem.

2 Telomeres prevent degradation of chromosomes by nucleases.

3 Telomeres prevent the ends of chromosomes from fusing via complementary base pairing.

4 The single stranded portion of the telomere serves as a template for extension via complementary base pairing.

A 1 only

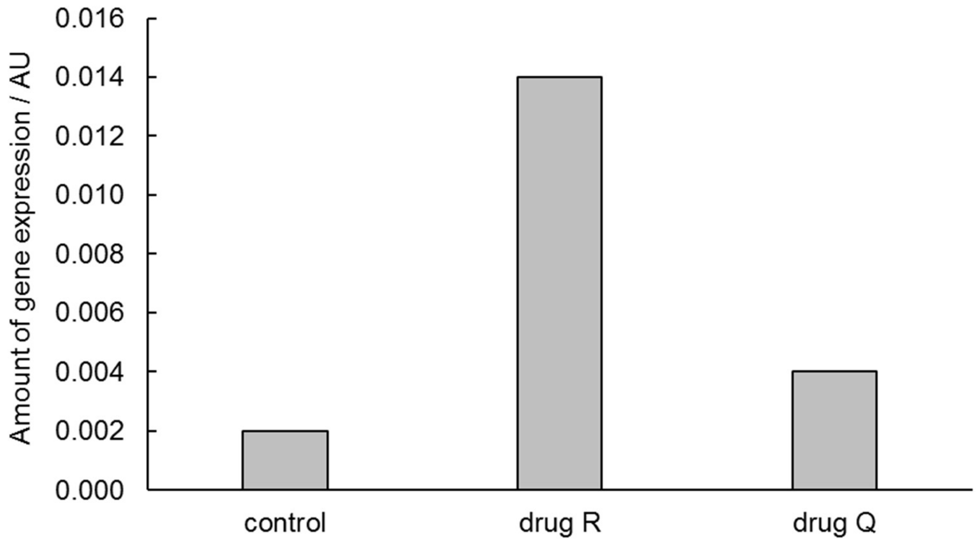
B 1 and 4 only

C 3 and 4 only

D 1, 2 and 4 only

15. Drug R is a DNA methyltransferase inhibitor and drug Q is a histone deacetylase inhibitor. An experiment was carried out to investigate the effects of drug R and Q on expression of a gene.

The graph shows the experimental results.



Based on the results shown which of the following could be true?

1 The control experiment shows the level of gene expression in the absence of both drugs.

2 Drug Q controls modification of histones, resulting in weaker binding of histones to DNA.

3 Drug Q increases gene expression by increasing accessibility of RNA polymerase to the promoter.

4 Drug R increases gene expression by preventing DNA methylation.

5 Drug R is less effective at inhibiting gene expression than drug Q.

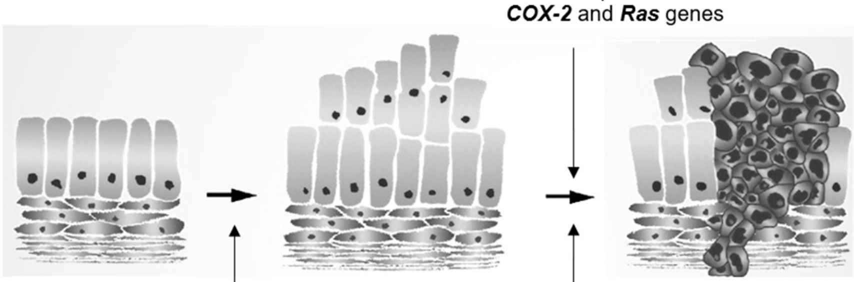
A 5 only

B 2, 3 and 4 only

C 1, 2, 3 and 4 only

D All of the above

16. The diagram below illustrates the development of colorectal cancers.



Based on the diagram as well as your understanding of how cancer develops, which of these statements can be inferred from this multistep model of carcinogenesis?

1 Cells whose *APC* and *β-catenin* genes are inactivated have lost contact inhibition and can form a tumour mass.

2 *APC* and *β-catenin* genes are tumour suppressor genes.

3 High levels of *Ras* protein are produced only when both copies of *Ras* gene are mutated.

4 Two copies of normal *p53* alleles must be present to inhibit cell division.

5 Gain-of-function mutation in *COX-2* gene increases the chances of metastatic carcinoma.

A 1 and 3 only

B 2, 3 and 4 only

C 1, 2 and 5 only

D 2, 3 and 5 only

17. Which of the following statements are correct?

1 Chlorophyll b is an accessory pigment that is involved in the resonance transfer of electrons from one pigment molecule to the next in the light harvesting complex of both photosystems I and II.

2 ATP synthase is located at the thylakoid membrane and the membrane of the intergranal lamella.

3 When reduced, NADP can donate electrons to the electron carriers in the electron transport chain on the thylakoid membrane.

4 The electrons released during the photolysis of water can be used to reduce NADP during non-cyclic photophosphorylation.

A 1 and 2 only

B 2 and 4 only

C 3 and 4 only

D 1 and 3 only

18. Some information about the respiratory enzymes that catalyse the reactions is shown in the table below.

enzyme	information
fructose 1,6-bisphosphate aldolase	• four identical subunits • changes to any one of the subunits means that the enzyme cannot function
hexokinase	• one subunit • active site changes shape to enclose the reactants
phosphofructokinase	• four identical subunits • has allosteric sites in addition to an active site
phosphoglucose isomerase	• two identical subunits • has a cytokine function when secreted into the external medium
pyruvate kinase	• four identical subunits • ATP acts as an inhibitor to regulate glycolysis
triosephosphate isomerase	• two identical subunits • each subunit has 14 alpha helices and 8 beta-pleated sheets

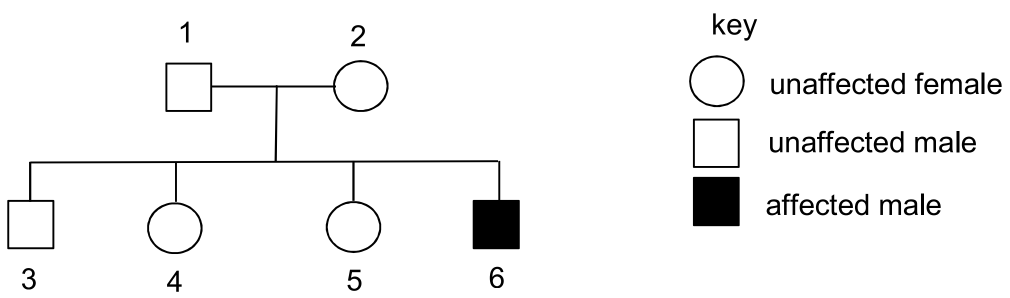
A student made the following deductions using the information provided in the table:

- Phosphoglucose isomerase, when secreted, can have a non-catalytic role.
- Only three of the six enzymes display quaternary protein structure.
- The active site of phosphofructokinase will change shape to allow the enzyme to regulate the rate of glycolysis.
- Each enzyme is coded for by one gene.
- The reaction catalysed by hexokinase is an induced-fit mechanism.

How many of the student's deductions are correct?

A 1 **B** 2 **C** 3 **D** 4

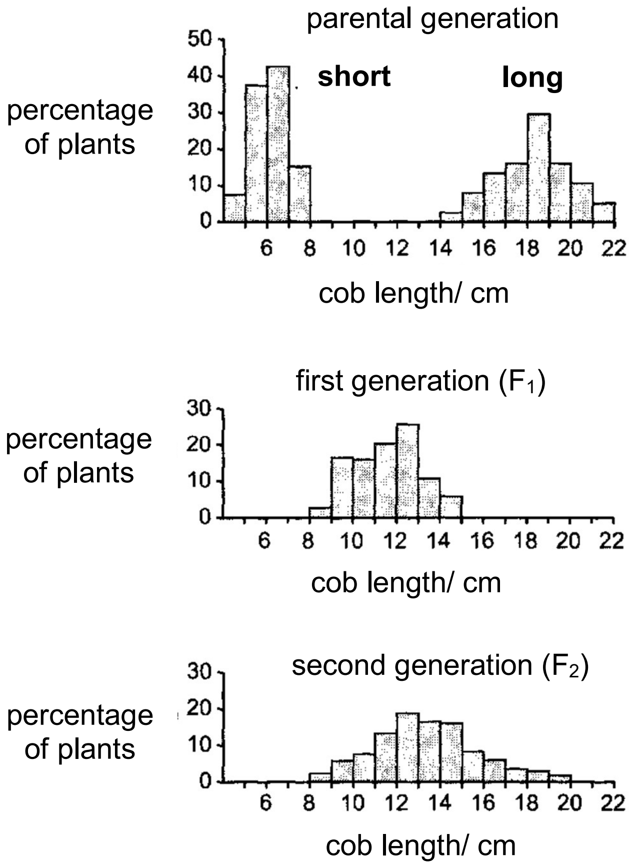
19. The diagram shows the inheritance of haemophilia in a family.



If individual 4 married a normal male, what is the probability that their first child would suffer from haemophilia?

A 0 **B** 0.125 **C** 0.25 **D** 0.5

20. The graphs show the distribution of measurements of cob length of corn kernel in three corn generations.



Which of the following is evidence that corn cob length is not controlled by a single pair of alleles showing incomplete dominance?

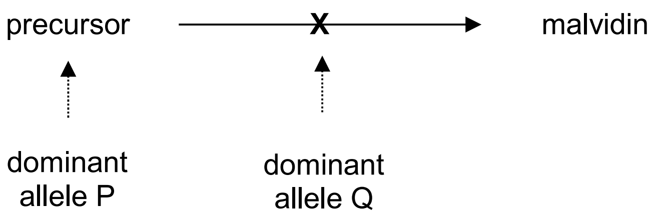
A The corn plants do not fall into distinctive, clear-cut groups.

B The phenotypic variation in each generation shows the influence of environmental factors.

C The F_2 generation does not occur in a 1: 2: 1 ratio.

D The F_1 generation is of intermediate corn cob length.

21. In the *Primula* plant, production of the pigment malvidin creates blue flowers. Synthesis of malvidin is controlled by a dominant allele **P** at the **P/p** gene locus which codes for a colourless precursor substance. However, the conversion of the precursor substance to malvidin can be suppressed by another dominant allele **Q** at the gene locus **Q/q** as shown in the biochemical pathway below. Flowers with the absence of malvidin will appear white.



When two plants with white flowers are crossed, the offspring generation has 44 plants with white flowers and 10 plants with blue flowers.

Which of the following statements is not true?

A The probability of obtaining a pure-breeding offspring at both loci with white flowers is 1 of 9.

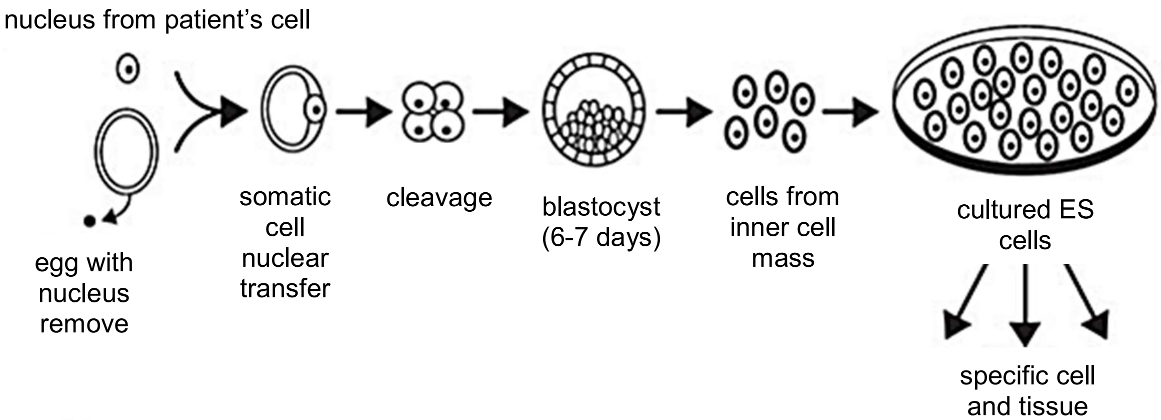
B The deviation from the Mendelian ratio (9:3:3:1) is due to gene interaction in which the expression of gene Q masks the phenotypic expression of gene P.

C The genotypes of the parents must be PpQq x PpQq.

D The genotypes PPQQ, PPQq, PpQq and ppQq produce the same phenotype.

22. The use of embryonic stem (ES) cells has encountered resistance on ethical grounds. To overcome this, researchers have come up with a way of developing embryonic stem cells from the cells of affected persons.

The figure below shows the theoretical process of development of these embryonic stem cells. The differentiated cell types would be introduced into the patient where needed.



Which of the following option(s) is/are true?

I The affected persons will not experience much of an immune response.

II Issues of informed consent are no longer relevant since the embryo is not donated.

III No embryo is destroyed in the process.

IV The argument that stem cell research violates the sanctity of life is not relevant here.

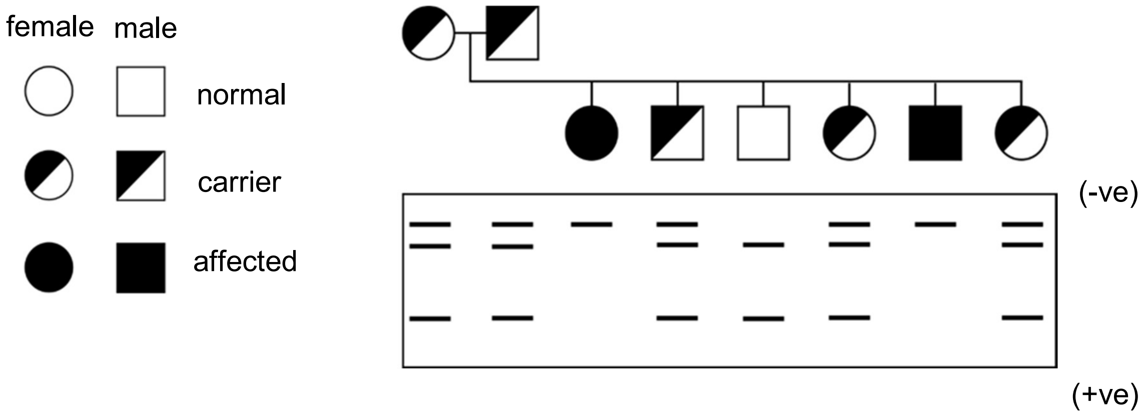
A I only

B I and II only

C II and III only

D All of the above

23. In order to understand the inheritance of disease X, DNA of members of a family were subject to polymerase chain reaction, gel electrophoresis and Southern blotting. The diagram below shows the autoradiograph obtained.



Which of the following statements can be concluded from the autoradiograph?

I The recessive allele is missing a restriction site.

II The binding site of the radioactive probe does not contain the restriction site.

III The disease is sex-linked recessive.

IV The nitrocellulose membrane was washed after nucleic acid hybridisation.

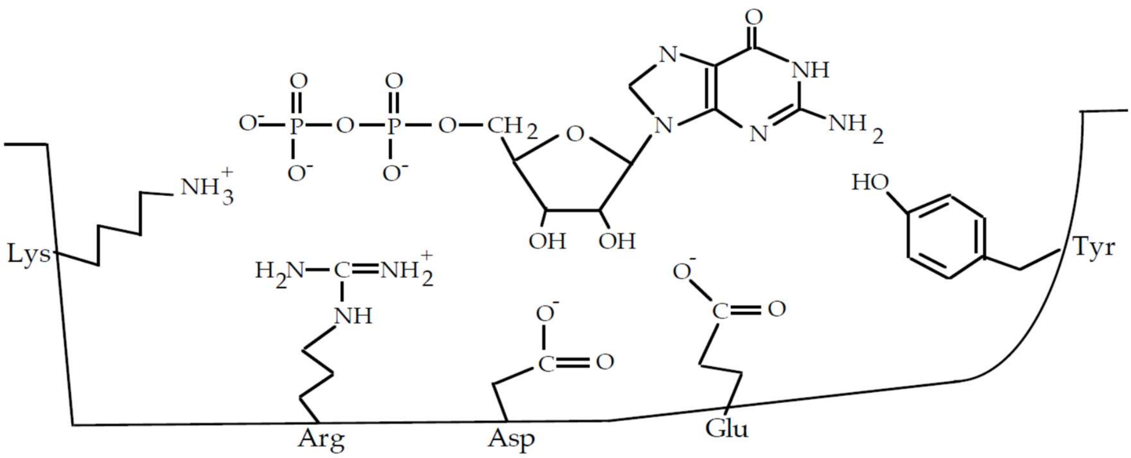
A I and III only

B I and IV only

C II and III only

D II and IV only

24. The figure below shows GDP in the binding pocket of a G protein.



Which of the following statement(s) is/are correct?

I GDP is attached to G protein because of complementary shape and charge.

II Phosphorylation of GDP to GTP makes the binding to G protein stronger, activating the G protein in the process.

III GDP is a deoxyribonucleoside diphosphate with guanine as the nitrogenous base.

IV The binding of GDP to the G protein is via weak interactions.

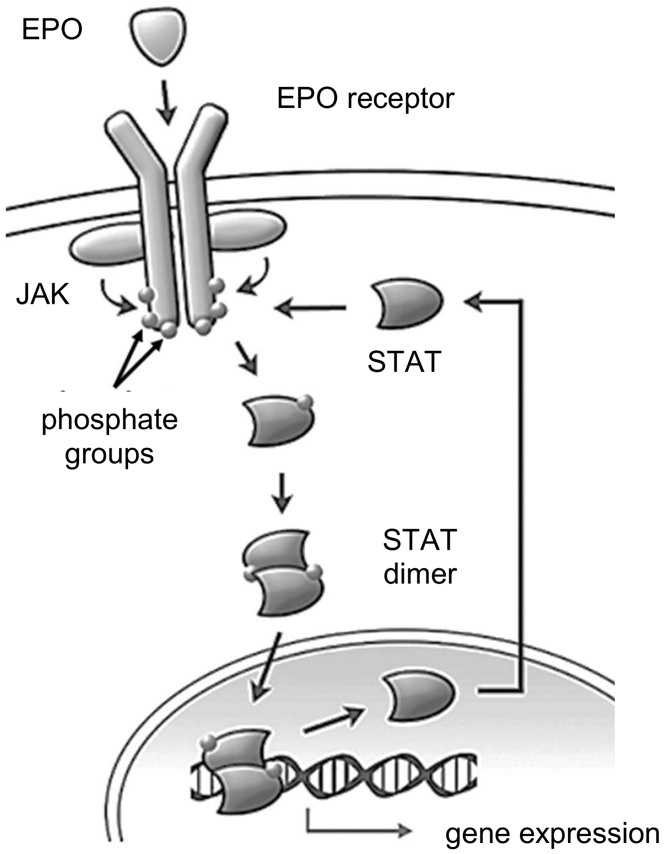
A I only

B I and IV only

C II and III only

D All of the above

25. The diagram shows the JAK-STAT cell signalling pathway.



Which of the following statements are correct?

1 EPO is a large and non-polar lipid hormone and hence it needs a cell surface receptor to cross the membrane.

2 Phosphorylation of STAT causes them to dimerize.

3 Gene expression is terminated when phosphatases remove phosphate groups from STAT dimers.

4 Signal amplification occurs as JAK phosphorylates multiple tyrosine residues on the EPO receptor.

A 1 and 3 only

B 2 and 3 only

C 2 and 4 only

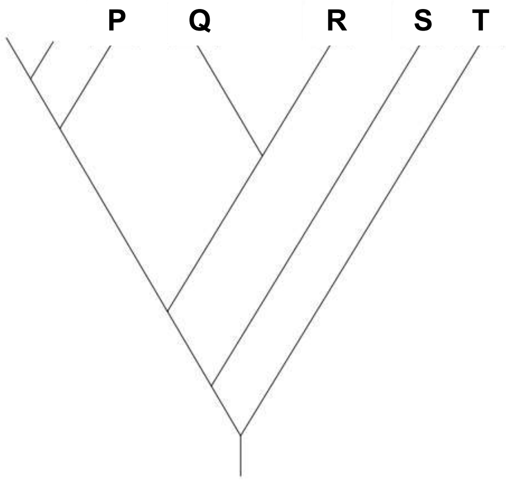
D 2, 3 and 4 only

26. Information about two species of penguin and their habitats is shown in the table below.

name of penguin species	photograph of penguin	mean adult body mass/kg	mean adult height/cm	endemic to	temperature range/ °C
Emperor penguin (*Apenodytes forsteri*)		34.0	115.0	Antarctica	-60 to -28
Galapagos penguin (*Spheniscus mendiculus*)		2.2	50.0	Galapagos Islands	19 to 31

Molecular phylogeny suggests that the emperor penguin split off from a branch which led to the evolution of all other living penguin species. This split happened 40 million years ago.

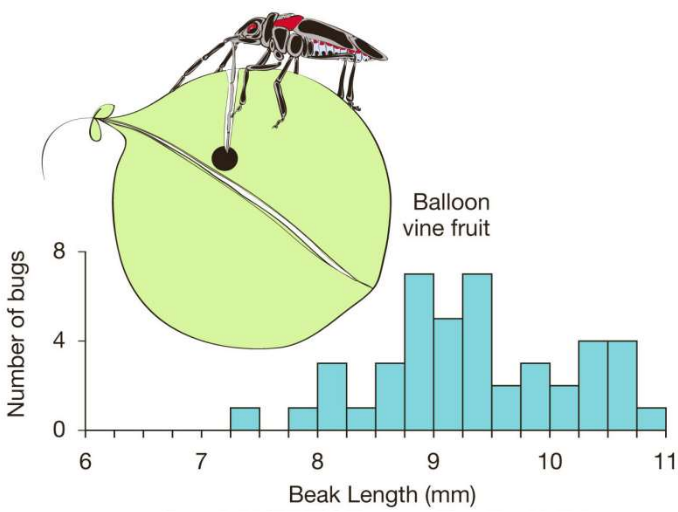
The diagram on the right indicates the evolutionary relationships between the main species of penguins. The letter Q indicates the position of the Galapagos penguin.

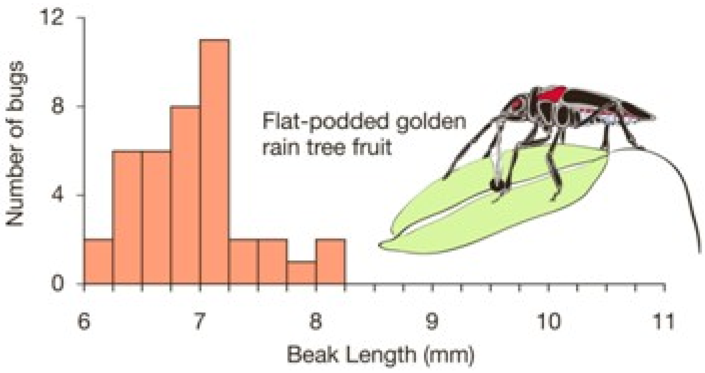


Based on the information given above, which of the following is correct?

	position of the emperor penguin on the diagram	position of the species most closely related to the Galapagos penguins on the diagram	anatomical adaptation of the emperor penguin to its habitat
A	S	P	having a large surface area to volume ratio to reduce heat loss
B	T	R	having a small surface area to volume ratio to reduce heat loss
C	R	S	reduced resting metabolic rate
D	P	R	thick layer of fat

27. Soapberry bugs in Florida originally fed on the native balloon vine, using their sharp beaks to penetrate the fruits to reach the seeds. In the 1920s, the flat-podded golden rain tree was introduced from Asia to mainland central Florida. This has thinner-skinned fruit, and correspondingly, soapberry bugs evolved shorter beaks after they switched completely to feed on this new host plant.





What best describes the type of evolutionary process depicted above?

The Soapberry bugs are undergoing

A genetic drift

B stabilising selection

C disruptive selection

D directional selection

28. Which statements about the use of antibiotics could cause an increase in antibiotic-resistant bacteria?

1 Patients do not always complete the course of antibiotics.

2 Patients keep unused antibiotics from previous prescriptions and take them at a later date in smaller doses than prescribed.

3 Patients do not take antibiotics.

4 Antibiotics are used in farming to prevent infections in livestock.

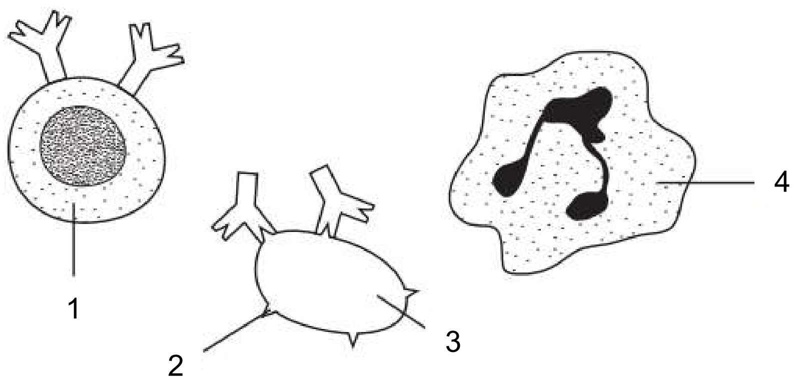
A 1,and 2 only

B 1 and 3 only

C 1, 2 and 4 only

D 3 and 4 only

29. The diagram shows different elements involved in the immune response in humans.



What are the parts labeled 1 to 4?

	1	2	3	4
A	lymphocyte	antigen	pathogen	phagocyte
B	phagocyte	antibody	pathogen	lymphocyte
C	phagocyte	antigen	lymphocyte	pathogen
D	pathogen	antigen	lymphocyte	phagocyte

30. What is the difference between T and B lymphocytes?

	T lymphocytes	B lymphocytes
A	do not form plasma cells	form plasma cells that secrete antibodies
B	does not stimulate macrophages to carry out phagocytosis	stimulate macrophages to carry out phagocytosis
C	formed from cells in the thymus	formed from cells in the bone marrow
D	produce memory cell	do not form memory cells

End of Paper